



Cellular signal processing used to study cancer cells

Purdue University researchers have reverse-engineered a cellular signal processing system and used it like a logic gate—a simple computer—to better understand what causes specific cells to migrate. Cancer cells are exposed to

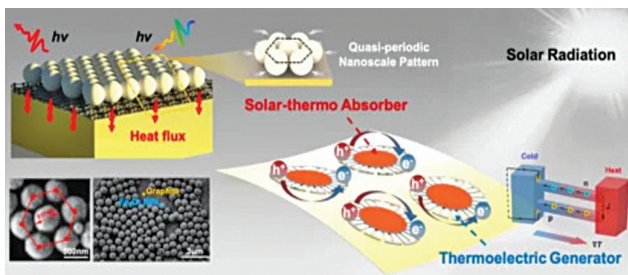


In microfluidic devices like this, Bumsoo Han's team observes how cancer cells behave in a simulated biological environment (Credit: Purdue University photo/Jared Pike)

multiple cues that cause them to move. The team adopted a ternary logic gate model to analyse these cues and predict how cells would move under different environments. Their experiments took place in a microfluidic platform with a centre chamber for the cells and two side platforms. Using this device, they could replicate fluidic flows in one direction, in the opposite direction, or no flow at all. They assigned values to which direction the cells moved under the two different stimuli. Based on observations, the cues of these cancer cells could now be diagrammed much like an electrical engineer would diagram a circuit.

A new solar harvester with enhanced energy conversion

Researchers from Harbin University, Zhejiang University, Changchun Institute of Optics, and the National University of Singapore have designed a new solar harvester with enhanced energy conversion capabilities. The new

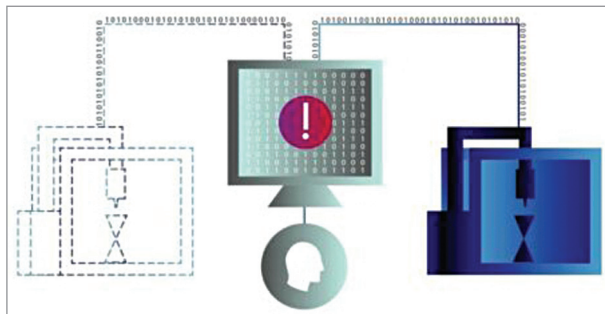


These images show the device's solar-thermal conversion (left) and solar thermoelectric harvesting (right) (Credit: Zifu Xu)

design aims to simplify fabrication and reduce costs while improving performance. It uses a quasiperiodic nanoscale pattern—meaning most of it is an alternating and consistent pattern, while the remaining portion contains random defects (unlike a nanofabricated structure) that do not affect its performance. Scientists built a flexible planar solar thermoelectric harvester, which reached a significant sustaining voltage of over 20 millivolts per square centimetre. They expect it to power 20 light-emitting diodes per square metre of solar irradiation. Researchers believe this strategy can serve low-power-density applications for more flexible and scalable engineering of solar energy harvesting.

Digital twin together with ML can now flag cyberattack indicators

A team of researchers at the National Institute of Standards and Technology (NIST) and the University of Michigan has devised a cybersecurity framework that brings digital twin technology together with machine learning and human expertise to flag indicators of cyberattacks. The team built a digital twin to emulate the 3D printing process and



A new and improved strategy for detecting cyberattacks on manufacturing systems, such as 3D printers, involves using AI to monitor a digital twin that mimics and is fed real-time data from the physical system (Credit: N. Hanacek/NIST)

provided it with information from the real printer. As the printer built a part, computer programs monitored and analysed continuous data streams, including both measured temperatures from the physical printing head and the simulated temperatures being computed in real time by the digital twin. In other words, the models were adept at recognising what the printer looked like under normal conditions, also meaning they could tell when things were out of the ordinary.

Believable tactile experience in VR made possible

Researchers from Rice University, Baylor College of Medicine, and Meta Reality Labs have found a hands-free way